# Agent-based Diamond–Dybvig with cultural heterogeneity

Walter Stover · George Mason University · 26 October 2026

# The puzzle: who runs?




*CryptoSlate, 12 October 2026, on the Financial Stability Board's note to the G20. Screenshot, cryptoslate.com.*

# Who runs? Luck of the draw, and location

## Variation in the signal

- Goldstein–Pauzner: every depositor gets a noisy draw of the same signal
- The draw breaks common knowledge and selects the equilibrium
- Who runs: whoever drew low
- Everyone decides at once, so there is no first

## Variation in location

- Networks and media decide who hears what, from whom, when
- Kelly–Ó Gráda, Iyer–Puri, Cookson et al.
- Who runs first: whoever sits nearest the source

# Variation in interpretation

**When there are two signals, which signal does a depositor act on, and in what proportion?**

**Silicon Valley Bank's shares are tanking as a mess unfolds**

**Venture firms are advising portfolio companies to move money out of SVB**

VCs are advising startups to generally keep no more than $250,000 in SVB checking accounts.

TechCrunch

Given recent banking-related carnage in the tech and tech-adjacent worlds, there's concern in the market that not all is well at SVB. Company CEO Greg Becker said in a call with venture clients that their assets are safe and that the stock sale was announced as an attempt to increase financial flexibility, strength and profitability at the bank.

Becker said the bank has "ample liquidity" to support its clients "with one exception: If everybody is telling each other that SVB is in trouble, that will be a challenge." The executive asked VC clients to "stay calm. That's my ask. We've been there for 40 years, supporting you, supporting the portfolio companies, supporting venture capitalists."

*TechCrunch, 9 March 2023. Screenshot, techcrunch.com.*

# Cultural economics: a source of stable weights

**Culture supplies a weight that differs across depositors, is set before the run, and does not move with the information environment.**

**Transmitted, not chosen.** Cultural values pass largely unchanged from parents to children (Guiso, Sapienza, Zingales; Bazzi et al.). The weight is a disposition, not a calculation.

**Individualism is the right dimension.** Self-reliance against deference to the group (Hofstede) is the private-against-social weighting the model already contains (Gorodnichenko–Roland).

**Measurable per depositor.** Given names carry it (Twenge et al.), so the weight can be observed, not assumed.

# The model: one weight on two signals

$$\hat{s}_i = \lambda_i \cdot y + (1 - \lambda_i) \cdot x_i$$

*the depositor's estimate of the bank, compared with a threshold: run if it falls below*

**$x_i$ what the depositor sees**

a private signal of the fundamental, one noisy draw each

**$y$ what the depositor is told**

a public signal, common to all, which can be stale

**$\lambda_i$ weight on the public line**

a disposition set before the run; culture is its source

**Individualists** $\lambda_I < \lambda_C$ **collectivists**. Same two signals, different estimates: $\hat{s}_I - \hat{s}_C = (\lambda_C - \lambda_I)(x - y)$. **Who runs is sorted by the weight, and the order flips with the sign of the public error.**

# The model in one picture

## Entities and state variables

**Depositors (N = 1,000)**

deposit $d_i \sim$ LogNormal(1, σ), capped at 1% of $D$
type I or C; weight $\lambda_I < \lambda_C$
private signal $x_i = \theta + s\,\varepsilon_i$
balance left; withdrawn or not

**Social network**

Newman–Watts small world (k, p): each depositor sees the share of its neighbours who have withdrawn

**Bank**

deposits $D$ in total; reserves $rD$ held as cash
long asset $(1 - r + e)D$; $e$ = equity per unit of $D$
sold at $1 - \delta$ per unit if cash runs short
returns $R(\theta) = 1 + 0.4\,\theta$ at period 2
promises $R_{stay}$ per unit left in

**Environment**

fundamental $\theta \sim$ U[0, 1], one per run
public signal $y$: none, stale ($y_0$),
tracks $\theta$, or corrects ($\rho$ per tick)

## Process schedule (one run)

**1 Cultural warm-up**

Flache–Macy dynamics on the network; the top μ become individualists (I), the rest collectivists (C)

**2 Draw the run's world**

$\theta$; each depositor's $x_i$; the public signal $y_1$

**3 Exogenous shock**

a random set of depositors withdraws everything

**4 Tick loop**

banked depositors, in random order

**a Observe**

neighbours' withdrawals; public $y_t$

**b Form beliefs (λ on others)**

$\theta$: $(1 - \lambda)\,x_i + \lambda\,y_t$
runs: $(1 - \lambda)$ own prior + $\lambda$ neighbours

**c Value by Monte Carlo**

stay: period-2 payoff on the balance left in
withdraw: a fraction f ∈ {0, ¼, ½, ¾, 1},
paid now, after any fire sale

**d Act**

withdraw the best f; if cash is short,
the bank sells the long asset at $1 - \delta$

↻ next tick, while anyone withdrew

**5 Period 2**

liquidity failure if cash ran out mid-run; else the bank fails if cash + long asset at $R(\theta)$ < promises

**6 Record**

failure type; who ran by type ($W_I$, $W_C$; D and D$);
timing by tick; size deciles; partial withdrawals

# Individualists and collectivists: who listens to whom

## Individualists

- Trust their own information about the bank
- Put little weight on what neighbours do or on public news
- Weight on the social side: $\lambda = 0.1$ to $0.3$
- Act on what they see themselves

## Collectivists

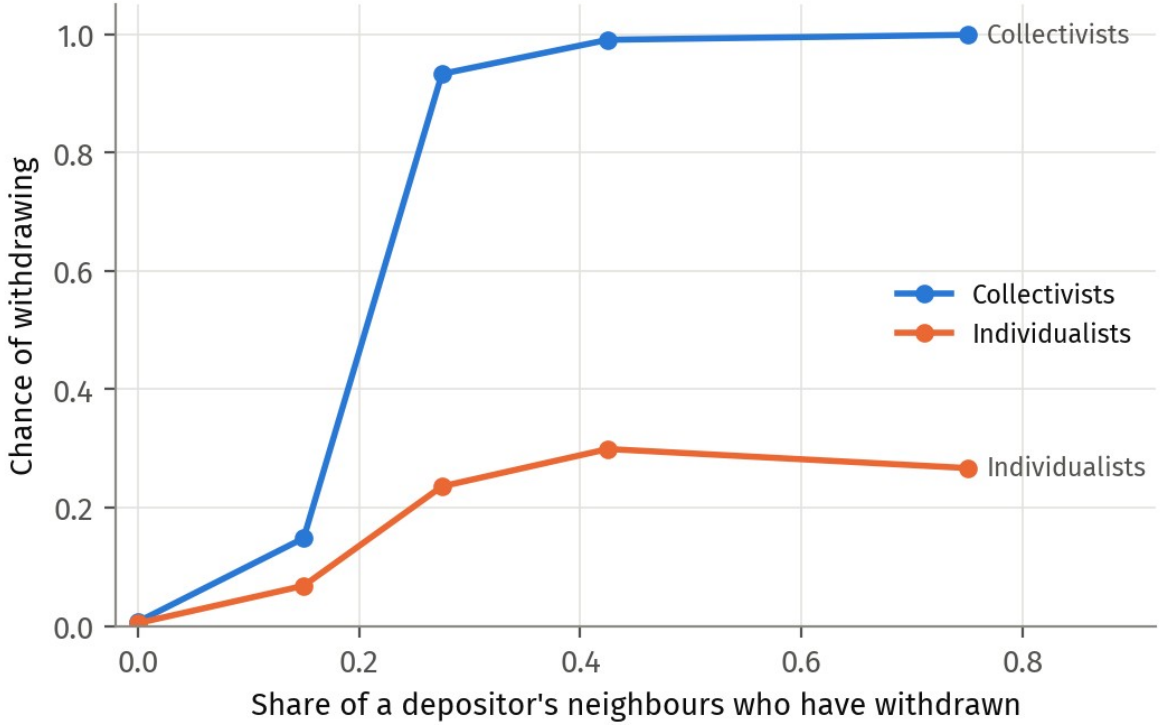
- Watch what neighbours do and what the public news says
- Put little weight on their own information
- Weight on the social side: $\lambda = 0.5$ to $0.9$
- Move when others move

**Three pairs ($\lambda$ individualist, $\lambda$ collectivist) in the figures**

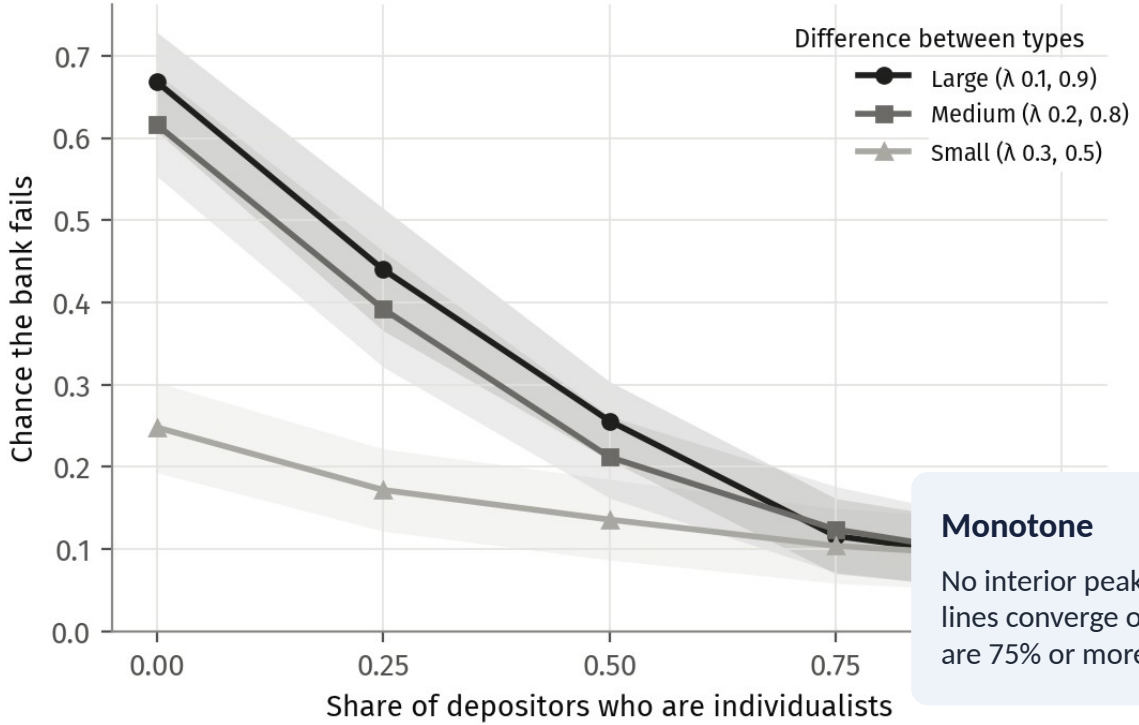
**(0.1, 0.9)** wide gap **(0.2, 0.8)** medium **(0.3, 0.5)** narrow

The share of individualists $\mu$ is set by a cultural warm-up on the network. **Orange = individualists**, **blue = collectivists** throughout.

# Collectivists herd



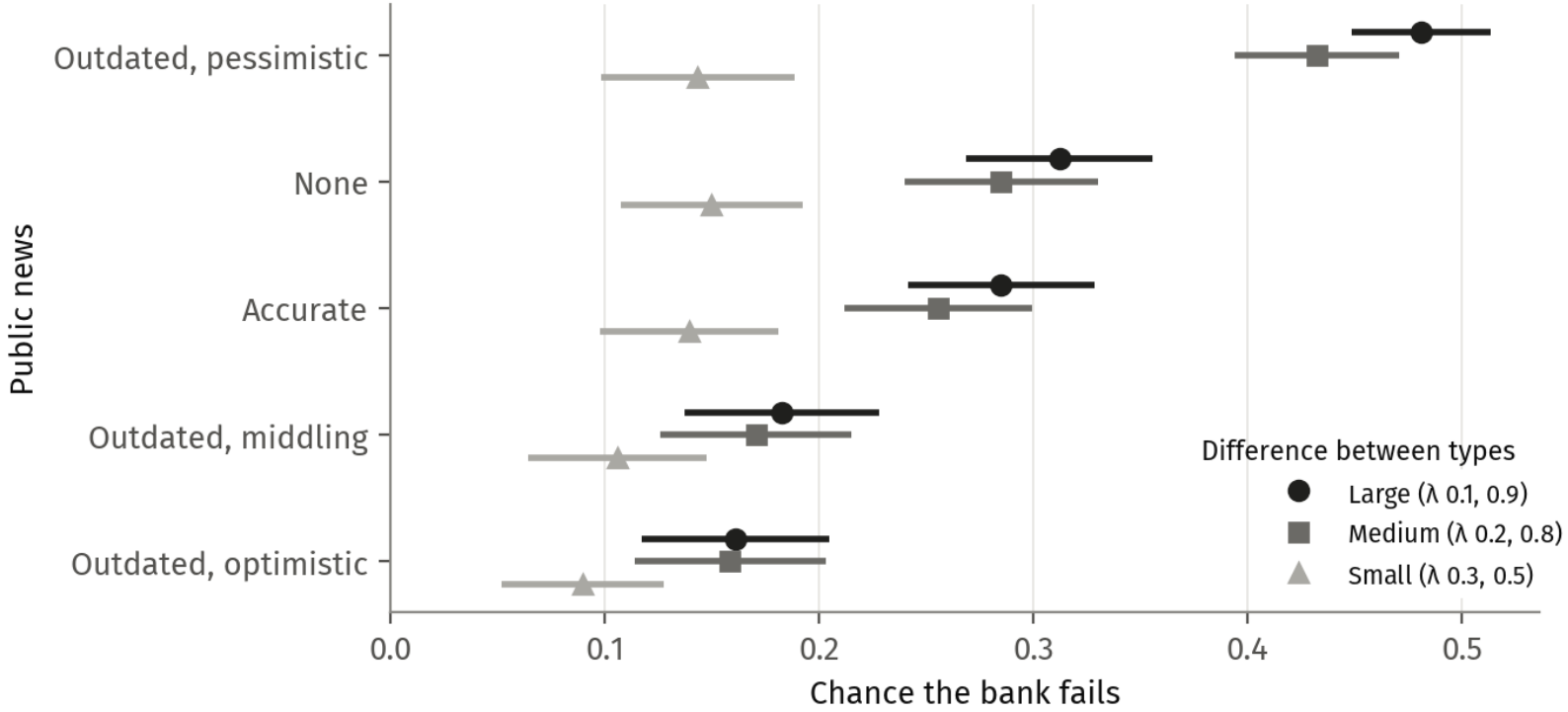
# More individualists, fewer failures



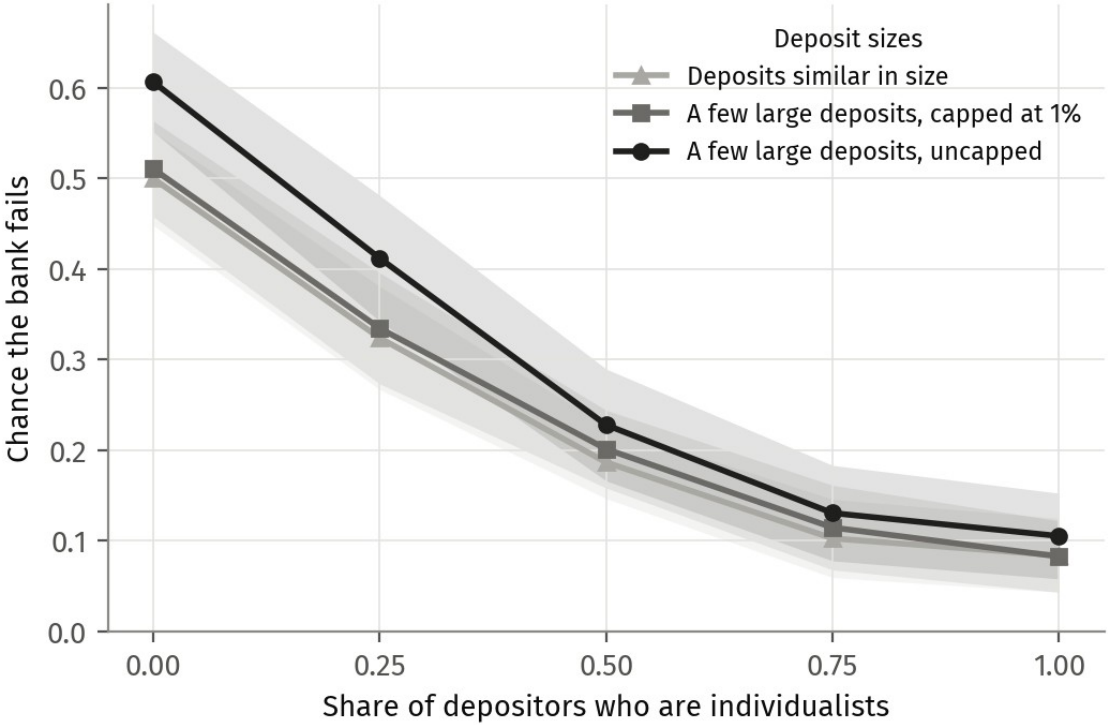

**Monotone**

No interior peak at any mix. The lines converge once individualists are 75% or more.

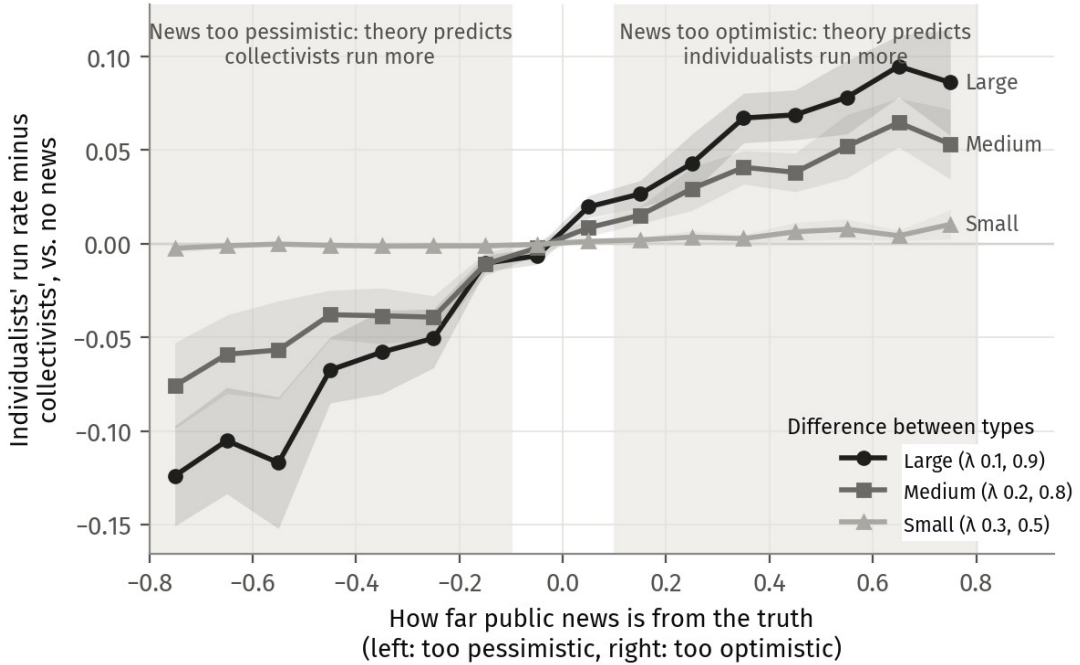
# Outdated news cuts both ways



# Varying the deposit-size distribution



# Does the model hold up?




*D relative to no news, against how far the public news sits from the truth. Shaded: where the theory predicts the sign.*

### Pre-registered tests

G1′ optimistic news: individualists run more +0.033

G4′ pessimistic news: collectivists run more −0.031

G2a′ the effect grows with the λ gap +0.093

G3 sorting with no news: counts against (one pair wrong-signed)

**3 of 4**

confirmatory tests support the theory, on all three versions of the bank

# Takeaways and next steps

| | | |
|---|---|---|
| 1 | **Sorting follows the news** | Optimistic public news makes individualists run relatively more; pessimistic news, collectivists. 3 of 4 pre-registered tests pass, in all three bank versions. |
| 2 | **Collectivists herd** | Their chance of withdrawing jumps from 0.15 to 0.93 once 20–35% of neighbours are out; individualists level off near 0.3. |
| 3 | **More individualists, fewer failures** | P(fail) falls from 0.67 to 0.08 as the individualist share goes from 0 to 1. Monotone; no interior peak. |
| 4 | **Stale news cuts both ways** | A stale optimistic signal halves failure, a stale pessimistic one raises it by half. Accurate news barely moves it. |
| 5 | **Capping the largest deposits** | A 1% cap brings failure down to the no-concentration bank's level; the culture gradient is unchanged. |
| 6 | **Not yet matched** | How Celsius's largest depositors exit: one-shot share 0.35 there, 0.93 or more in the model. |

## Next steps

- Super-individuals, to match how the largest depositors exited Celsius
- Deposit insurance and the reserve ratio on the fire-sale bank
- Network topology and learning
- Full sweep: stale public signal plus no-signal baseline, CARA 0.1, σ reported